Nanyang Technological University
School of Electrical & Electronic Engineering
IE2106 Engineering Mathematics I

Tutorial 6 Probability & Statistics

1. In the past it has been found that the standard deviation of diameter of holes drilled by a certain machine is 0.04 mm. The machine has been reset slightly, but there is no reason to believe the standard deviation has changed. 100 holes drilled by the machine are chosen at random and found to have a mean diameter of 5.2 mm. Give an interval estimate for the mean diameter of all holes with this setting, with 95% confidence that your estimate includes the true mean.

2. Depth perception is very important in dental work. A certain depth perception aptitude test asks subject to estimate the distance between a fixed object and a second object of variable position. A random sample of 14 dentists took the test and gave the following errors in their estimation (in mm):

1.1	1.5	0.9	2.1	1.4	1.7	0.8
1.3	1.8	1.1	1.6	1.9	1.2	1.6

   (a) Calculate the sample mean and sample standard deviation.
   (b) Find a 90% confidence interval for the population mean of errors for this depth perception test. State any assumptions needed.

3. A university admission officer is interested in estimating the variation of GPA (grade point average) of students in the Engineering Faculty who had similar results when they applied for admission. A random sample of 20 students revealed the following GPA's:

2.28	2.62	3.04	2.81	2.82	2.50	2.86	2.88	3.22	2.68
3.00	2.50	3.28	2.79	2.64	2.49	3.15	2.84	3.00	2.63

   The admission officer knows that the distribution of GPA's for such students is normal. Determine the 95% confidence interval for the population variance of GPA's of all such students.

Ans

1. $[5.19216, 5.20784]$
2. $[1.249, 1.611]$
3. $[0.04, 0.149]$